SLAM & Bayesian estimation

Week 12 · Where am I, and what surrounds me?

Estimate the robot

Bayesian beliefs
Particles and Gaussian filters

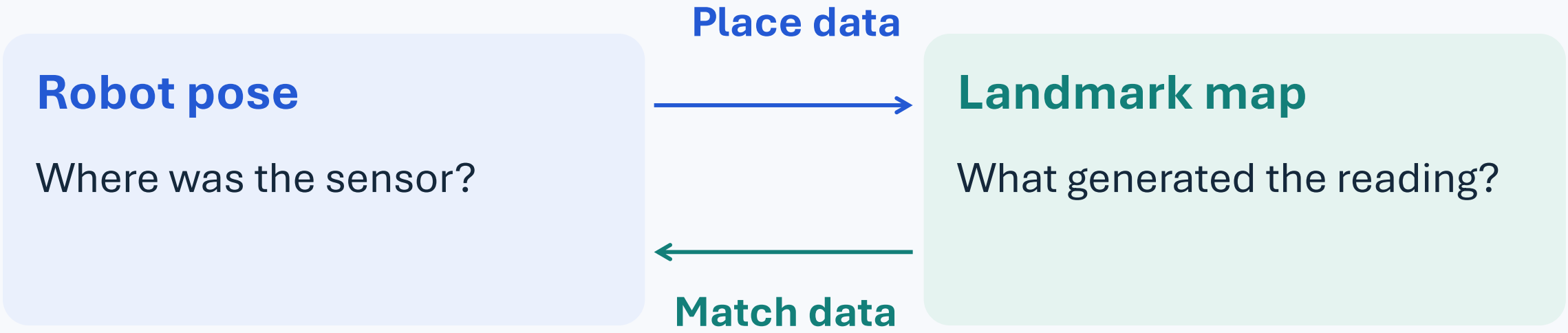
Estimate the map

Range-only landmarks
Joint uncertainty

30 min mathematics · 12 min integrated demo · 3 min recap

Why SLAM?

An unfamiliar environment gives us two unknowns.



Estimate the robot and its surroundings together.

Localization, mapping, SLAM

The distinction is what is given and what is estimated.

Localization

Given: map
Estimate: robot

Mapping

Given: poses
Estimate: map

SLAM

Estimate both
Keep their coupling

Fixed landmarks can still have unknown positions.

Define the symbols

A pose is more than a position; a map contains many landmarks.

$$x_t = (x, y, \theta); \quad m = \{m_1, \ldots, m_J\}$$

u_t: command; z_t: observation; c_t: landmark ID

t: time · j: landmark

i: particle · m: map

Demo state: p_t = (x,y); heading is fixed.

What does SLAM estimate?

A joint belief, rather than two independent guesses.

$$p(x_{0:t}, m \mid u_{1:t}, z_{1:t}, c_{1:t})$$

Online: $p(x_t, m \mid u_{1:t}, z_{1:t}, c_{1:t})$

Robot and map uncertainty are correlated.

The SLAM process

Use motion and observations repeatedly, maintaining uncertainty.

Predict → **Observe** → **Associate** → **Update**

Joint estimate

Robot belief + landmark map + uncertainty

Move → predict → observe → associate → update.

Bayesian prediction & correction

Week 10 beliefs, now over continuous robot states.

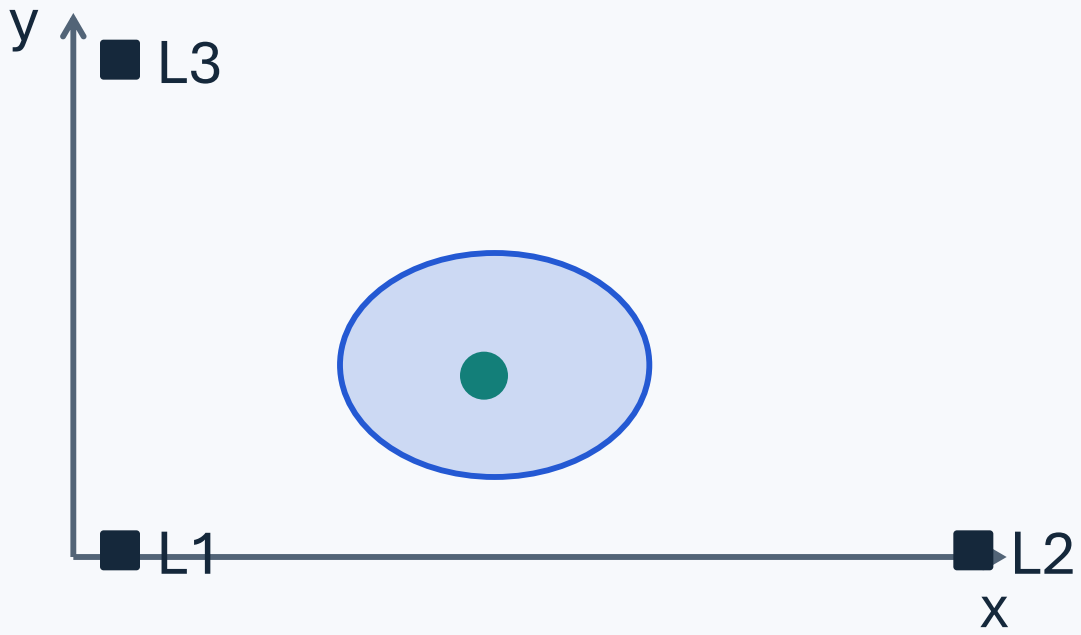
$$\bar{b}_t(x) = \int p(x \mid x', u_t)\, b_{t-1}(x')\, dx'$$

$$b_t(x) = \eta\, p(z_t \mid x, m)\, \bar{b}_t(x)$$

Prediction spreads possibilities; evidence changes plausibility.

Our range-only visualization

Same scene and sensor; different knowledge about the map.



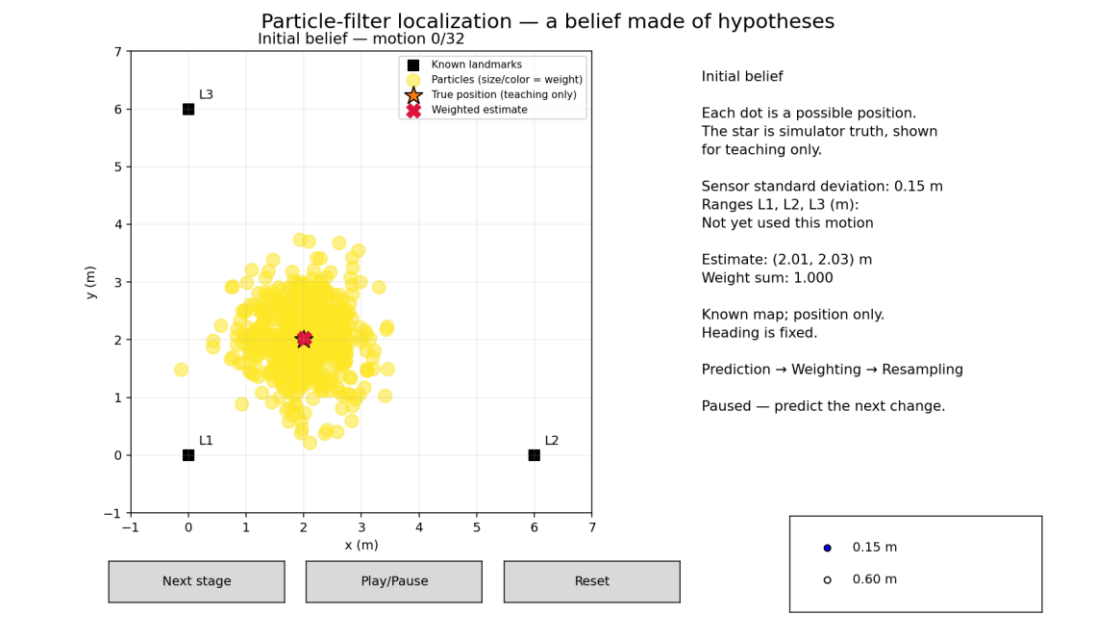
Model

500 robot particles
Three identified ranges
Fixed heading
Two noise presets

Localization knows coordinates; SLAM knows only IDs.

Live: inspect the initial belief

Localization mode · What would the real robot know?



Predict → reveal

Which information is known?

Known landmarks + uncertain robot position.

A belief made of particles

A weighted set of hypotheses approximates a distribution.



$$b_t(x) \approx \sum_i w_t^{(i)}\, \delta(x - x_t^{(i)})$$

$$\sum_i w_t^{(i)} = 1; \quad \hat{p}_t = \sum_i w_t^{(i)} p_t^{(i)}$$

A mean is a summary, not the whole belief.

Predict with the motion model

Move every robot hypothesis with its own sampled uncertainty.

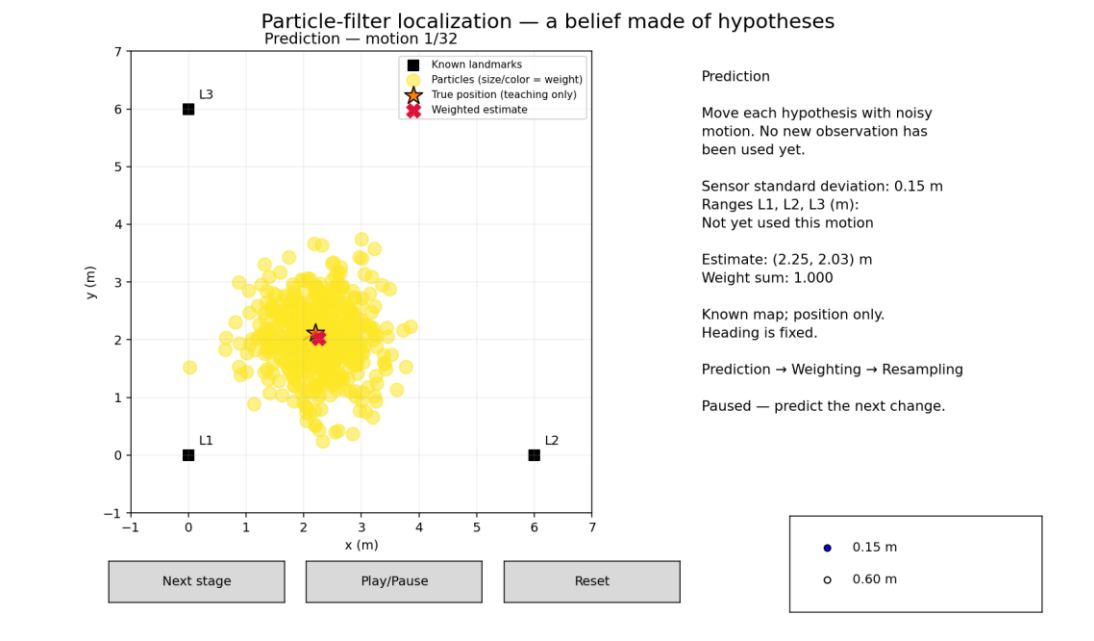
$$p_t^{(i)} = p_{t-1}^{(i)} + u_t + \varepsilon_t^{(i)}$$

$$\varepsilon_t^{(i)} \sim N(0, 0.08^2\, I)$$

No new measurement is used yet.

Live: predict before pressing

Next stage → Prediction



Predict → reveal

What moves?
What stays fixed?

What moves? What information is still missing?

Predict ranges; score agreement

Likelihood rewards hypotheses that explain the measurement.

$$h_j(p) = \|m_j - p\|_2; \quad z_{t,j} = h_j(p_t) + \nu_{t,j}$$

$$\nu_{t,j} \sim N(0, \sigma_z^2)$$

$$L_i \propto \exp[-\tfrac{1}{2} \sum_j ((z_{t,j} - h_j(p_t^{(i)}))/\sigma_z)^2]$$

σ_z = 0.15 m or 0.60 m.

Weight, then normalize

Keep particle positions; change their relative plausibility.

$$\tilde{w}_i = w_{t-1}^{(i)} \, L_i$$

$$w_t^{(i)} = \tilde{w}_i \, / \, \sum_k \tilde{w}_k$$

The estimate can move while the dots stay fixed.

Calculate three posterior weights

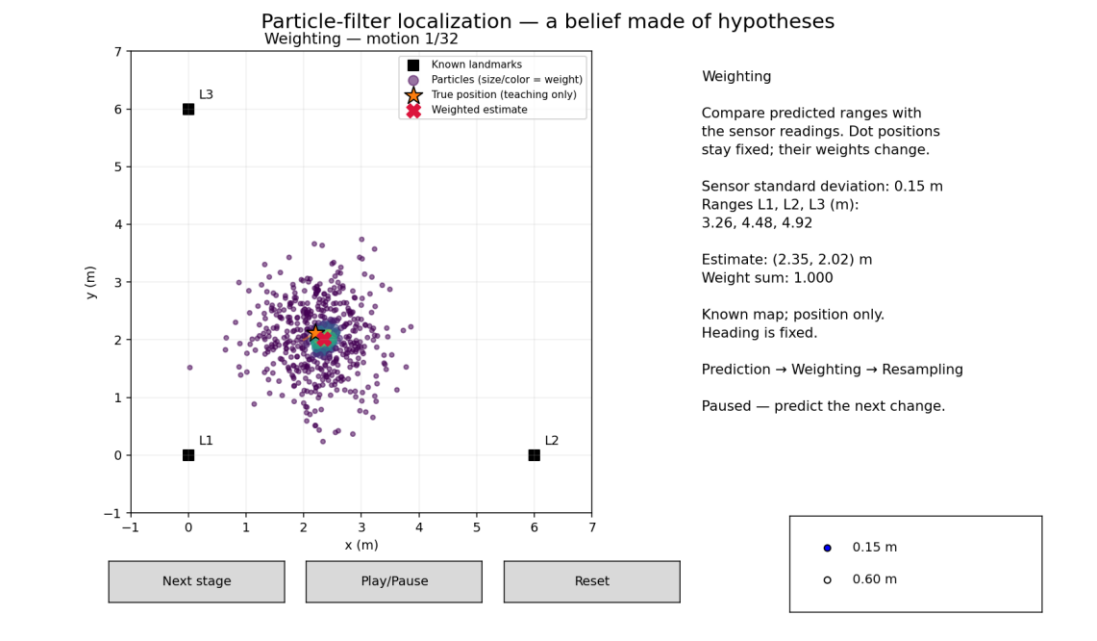
Equal priors · likelihoods 0.6, 0.3, 0.1

Particle	Prior × likelihood	Posterior	Copies / 10
A	(1/3) × 0.6 = 0.2	0.6	6
B	(1/3) × 0.3 = 0.1	0.3	3
C	(1/3) × 0.1 = 1/30	0.1	1

Normalizer = 1/3; posterior weights sum to 1.

Live: reveal weighting

Next stage → Weighting



Predict → reveal

Dots or weights?
Why does the mean move?

Why does the red estimate move?

Resample plausible hypotheses

Copy by probability; reset weights after selection.

Expected copies: 6 blue · 3 teal · 1 amber

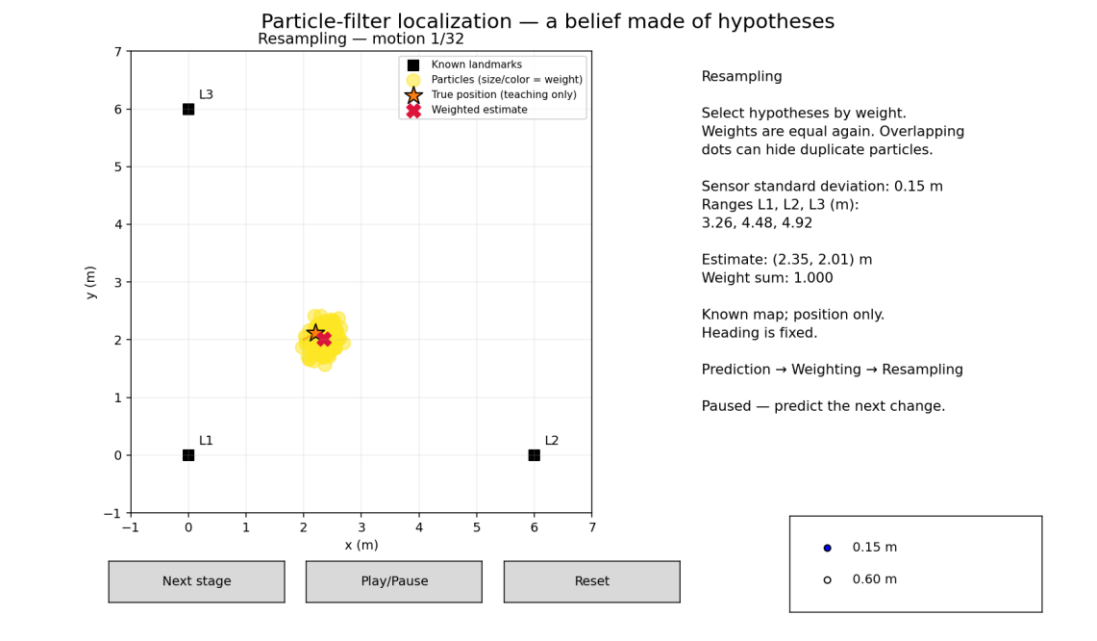


$$P(\text{select } i) = w_i; \quad \text{new weights} = 1/N$$

SLAM must copy the robot AND its conditional map.

Live: reveal resampling

Next stage → Resampling



Predict → reveal

Another measurement?
What gets copied?

Uniform weights do not mean uniform spatial belief.

Uncertainty and failure

A tight cloud can still be wrong.

Evidence uncertainty

Noisy readings
Ambiguous geometry
Incorrect noise model

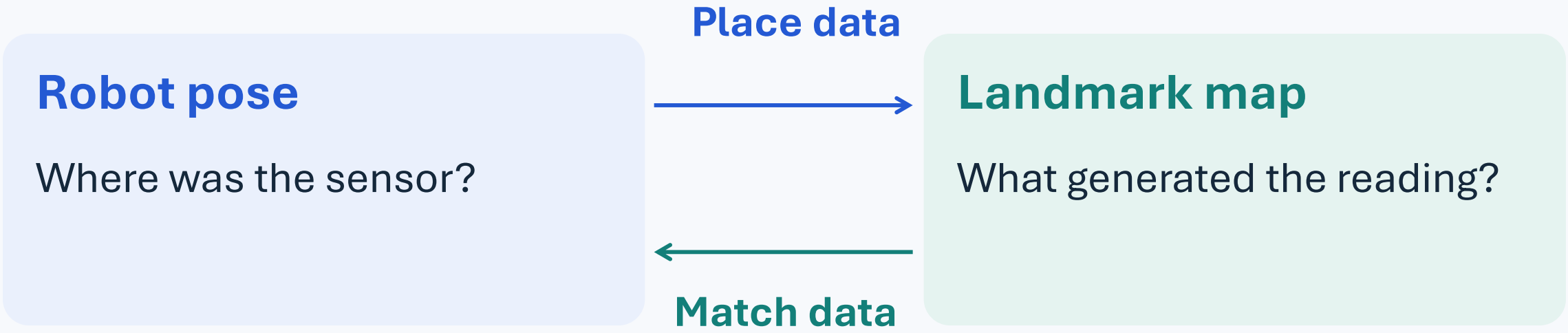
Approximation uncertainty

Finite robot hypotheses
Resampling diversity
Bounded landmark grids

Model mismatch · missing hypotheses · lost diversity

When the map is unknown

The same range now constrains robot AND landmark locations.

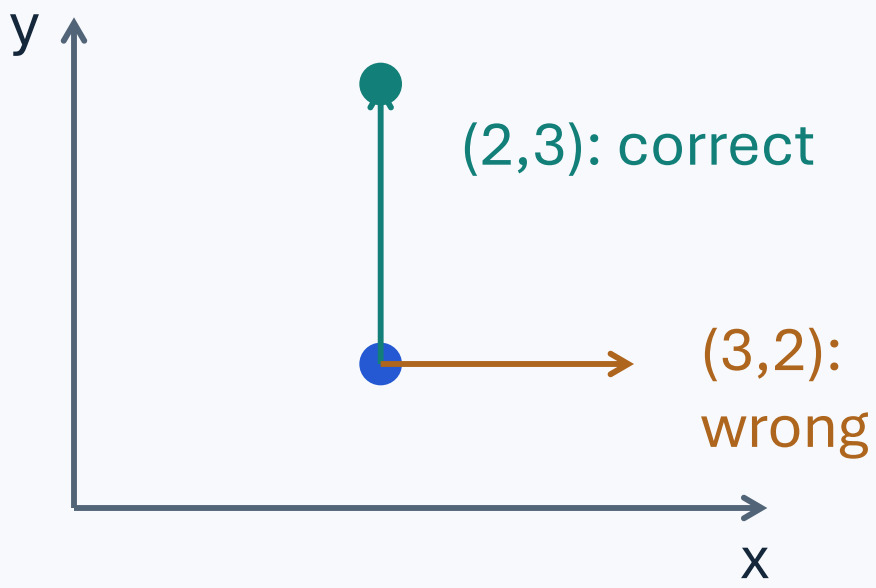


$$z_{t,j} \approx \|m_j - p_t\|_2$$

Static landmarks; unknown coordinates.

Pose errors distort the map

Frame mathematics explains the coupling.



$$m_j = t_t + R(\theta_t)\, z_j^{local}$$

Local point (1,0)
Robot (2,2), heading 90°

R = rotation; t = translation. Example needs direction information.

Which landmark was observed?

Data association connects observations to map features.

Match

An existing landmark
Use its map belief

Initialize

A new landmark
Keep uncertainty

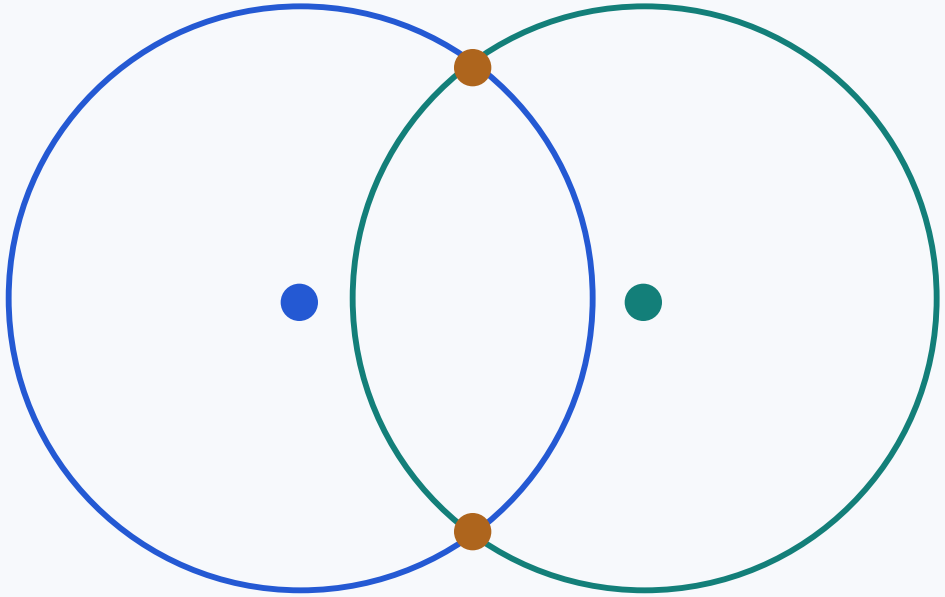
Reject / defer

Ambiguous match
Avoid false certainty

Demo IDs are supplied; real systems must often infer them.

Range-only landmark estimation

One reading gives a circle; new viewpoints narrow the possibilities.



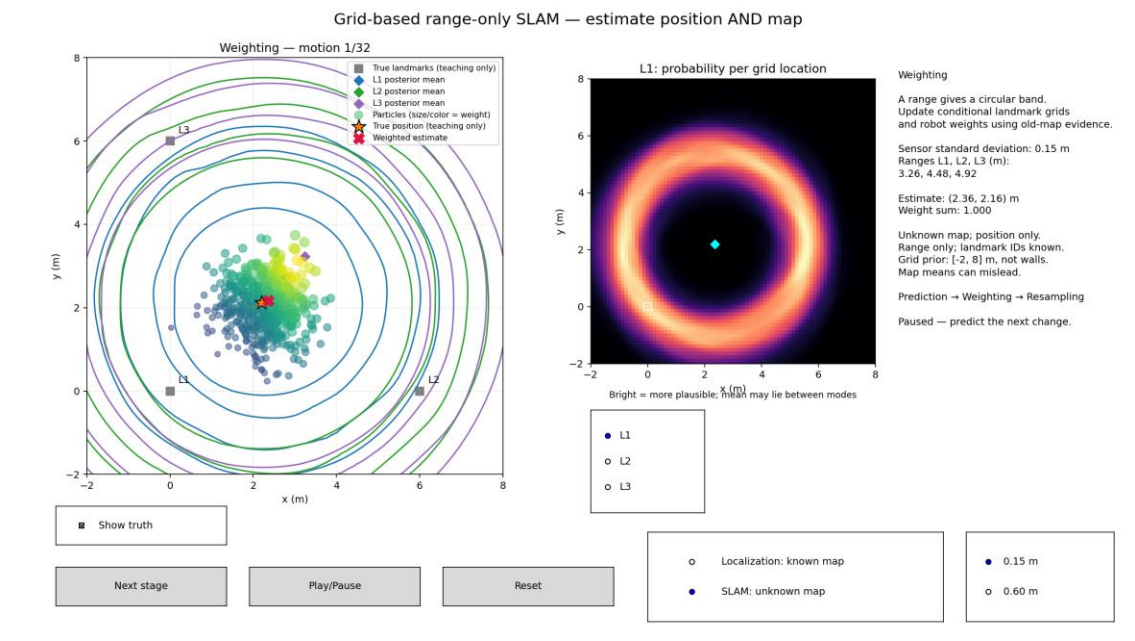
One range: a band
Two views: ambiguity

$$\|m_j - p_t\|_2 \approx z_{t,j}$$

Mean ≠ most likely landmark location.

Live: switch to range-only SLAM

SLAM: unknown map · Same physical landmarks and sensor



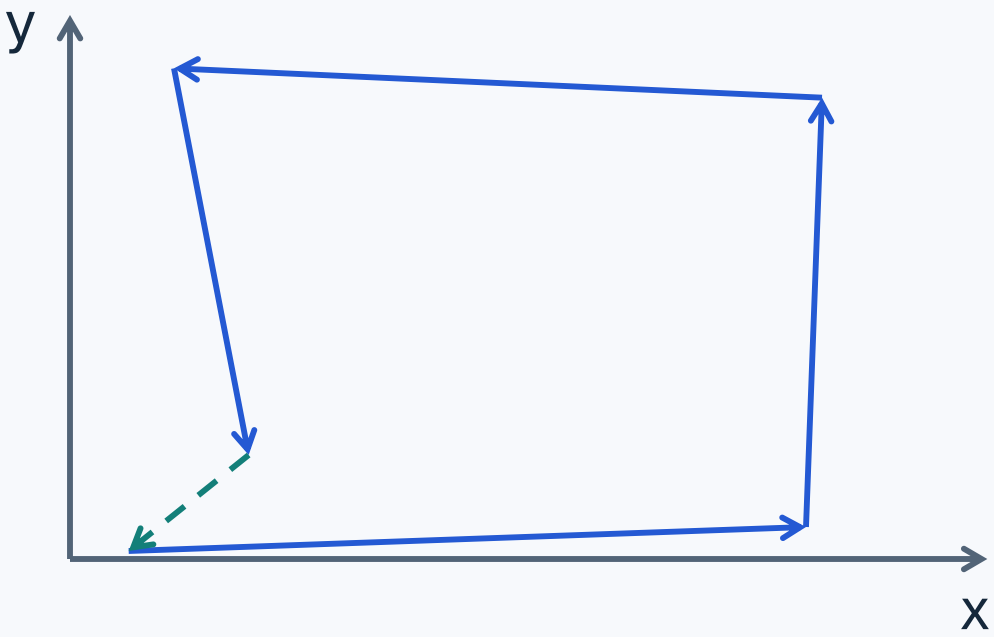
Change modes

Unknown locations
Range-only bands
Multiple viewpoints

A circle → overlapping constraints → a more informative map.

Drift and recognized revisits

Revisiting a place is useful only when the correspondence is recognized.



Recognized revisit

Adds a consistency constraint
Requires correct matching
Can inform correlated poses

Reobservation is implemented; global loop optimization is not.

Occupancy-grid mapping

A different map representation: free, occupied, or unknown.

$$\ell_t(c) = \ell_{t-1}(c) + \text{logit}\, P(c \text{ occupied} \mid z_t, x_t) - \ell_0(c)$$

$$P(c \text{ occupied}) = 1 / (1 + \exp(-\ell_t(c)))$$

HW6 connection; occupancy mapping is not implemented here.

Each robot hypothesis carries a map

Grid-based range-only SLAM uses conditional landmark distributions.

Particle i

Robot position
L1 distribution
L2 distribution
L3 distribution

$$e_{ij} = \sum_g L_{ij}(g)\, M_{t-1,ij}(g)$$

$$M_{t,ij}(g) \propto L_{ij}(g)\, M_{t-1,ij}(g)$$

Evidence first; posterior map second; copy them together.

SLAM does not require particles

Choose a representation suited to the model and uncertainty.

Particles

Sampled hypotheses
Multiple modes
Sampling cost

EKF / UKF

Mean + covariance
Nonlinear approximations
Correlated state

Graph SLAM

Pose/map constraints
Optimize residuals
Different architecture

Particle filters · Gaussian filters · graph optimization

The Kalman-filter idea

Represent uncertainty with a mean and covariance.

$$\bar{\mu} = F\mu + Bu; \quad \Sigma = F\Sigma F^T + Q$$

$$e = z - H\bar{\mu}; \quad S = H\Sigma H^T + C_z$$

$$K = \Sigma H^T S^{-1}; \quad \mu^+ = \bar{\mu} + Ke$$

Linear Gaussian assumptions make this update exact.

EKF-SLAM: linearize the models

One Gaussian estimates robot pose and every landmark together.

$$X = [x, y, \theta, m_{1x}, m_{1y}, \ldots]^T$$

$$\bar{\mu} = f(\mu, u); \quad \bar{\Sigma} = F\Sigma F^T + Q$$

$$e = z - h(\bar{\mu}); \quad \mu^+ = \bar{\mu} + Ke$$

Jacobians approximate nonlinear motion and sensing.

Why cross-covariance matters

A landmark observation can inform the robot and the rest of the map.

$$\begin{bmatrix} \Sigma_{xx} & \Sigma_{xm} \\ \Sigma_{mx} & \Sigma_{mm} \end{bmatrix}$$

Shared uncertainty

Robot ↔ landmarks
Landmark ↔ landmark
Use the full covariance

UKF uses sigma points; correlations still matter.

Reconstruct the complete process

Different representations, the same estimation question.

Predict → **Observe** → **Associate** → **Update**

Joint estimate

Robot belief + landmark map + uncertainty

Predict → observe → associate → update robot AND map.

Questions & recap

Explain the evidence, the uncertainty, and what each mode estimates.

Explain the updates

Prediction versus evidence
Likelihood versus weight
Resampling versus sensing

Explain SLAM

Known versus unknown map
Pose–map coupling
Particles versus Gaussian filters

OPTIONAL / DEEPER DISCUSSION

Joint posterior and reference frame

Conditional independence requires explicit assumptions.

$$p(x_{0:t}, m \mid u, z, c) \propto p(x_0)\, p(m)$$

$$\times \prod_k p(x_k \mid x_{k-1}, u_k) \prod_j p(z_{k,j} \mid x_k, m_j)$$

$$p(m \mid x_{0:t}, z, c) = \prod_j p(m_j \mid x_{0:t}, z, c)$$

Online marginals differ from a stored full-trajectory posterior.

OPTIONAL / DEEPER DISCUSSION

Likelihoods and systematic selection

Stable arithmetic and correct ancestry preserve the model.

$$\log L_i = \text{const} - \tfrac{1}{2} \sum_j \left((z_j - h_j(p_i)) / \sigma_z \right)^2$$

$$u_k = (u_0 + k)/N; \quad u_0 \sim \text{Uniform}(0,1)$$

$$N_{\text{eff}} = 1 \,/\, \sum_i w_i^2$$

OPTIONAL / DEEPER DISCUSSION

Range-only conditional-map update

The evidence uses the previous map distribution.

$$e_{ij} = \sum_g L_{ij}(g)\, M_{old,ij}(g)$$

$$M_{new,ij}(g) = L_{ij}(g)\, M_{old,ij}(g) \,/\, e_{ij}$$

$$w_i^{+} \propto w_i \prod_j e_{ij}$$

$$P(m_j{=}g \mid z,u) \approx \sum_i w_i\, M_{ij}(g)$$

OPTIONAL / DEEPER DISCUSSION

Kalman/EKF covariance and checks

Covariance correction must preserve the joint structure.

$$\Sigma^{+} = (I - KH)\,\Sigma\,(I - KH)^{T} + K\,C_z\,K^{T}$$

$$F = \partial f/\partial X; \quad H = \partial h/\partial X$$

$$\rho_j = \|m_j - p\|; \quad \partial\rho/\partial p = -(m_j - p)^{T}/\rho_j$$

OPTIONAL / DEEPER DISCUSSION

Maps and graph-based SLAM

Landmark grids, occupancy cells, and factor graphs are different.

Occupancy: $\ell^{+} = \ell + \mathrm{logit}(p_{\mathrm{hit}}) - \ell_0$

Graph: minimize $\sum_f r_f(X)^{T}\, \Omega_f\, r_f(X)$